

Sales Performance Dashboard

Visual Report & Data Storytelling

Data Analyst Internship — Task 2: Data Visualization and Storytelling

Tool Used	Microsoft Power BI
Dataset	Sales / Superstore-style dataset (sales_data)
Objective	Create visualizations that convey a compelling business story
Deliverable	Visual report with dashboard screenshots and insights

Executive Summary

This report presents a Power BI dashboard built to analyze sales performance across products, countries, customer age groups, and time. The dashboard combines KPI cards, trend lines, a gauge, a donut chart, bar charts, a treemap, and a stacked bar chart to give both a high-level performance snapshot and the ability to drill into specific segments.

The headline story is one of a business that is **profitable but falling short of its growth targets**: revenue and order volume are both trailing their targets by wide margins, even though profit in absolute terms looks healthy on the gauge. Sales are concentrated in two countries and one age segment, and a small set of products (led by Tires and Tubes, Bottles and Cages, and Road Bikes) drive most of the volume.

Key Numbers at a Glance

Metric	Actual	Target	Gap
Revenue	4,134.99K	15,000K	-72.43%
Order Quantity	0.82K	1K	-17.8%
Profit	16M	up to 33M (scale)	positive, but below top of range

Full Dashboard View

The screenshot below shows the complete Power BI dashboard as built for this task, combining KPI cards, trend, distribution, geographic, and product-level visuals on a single page.

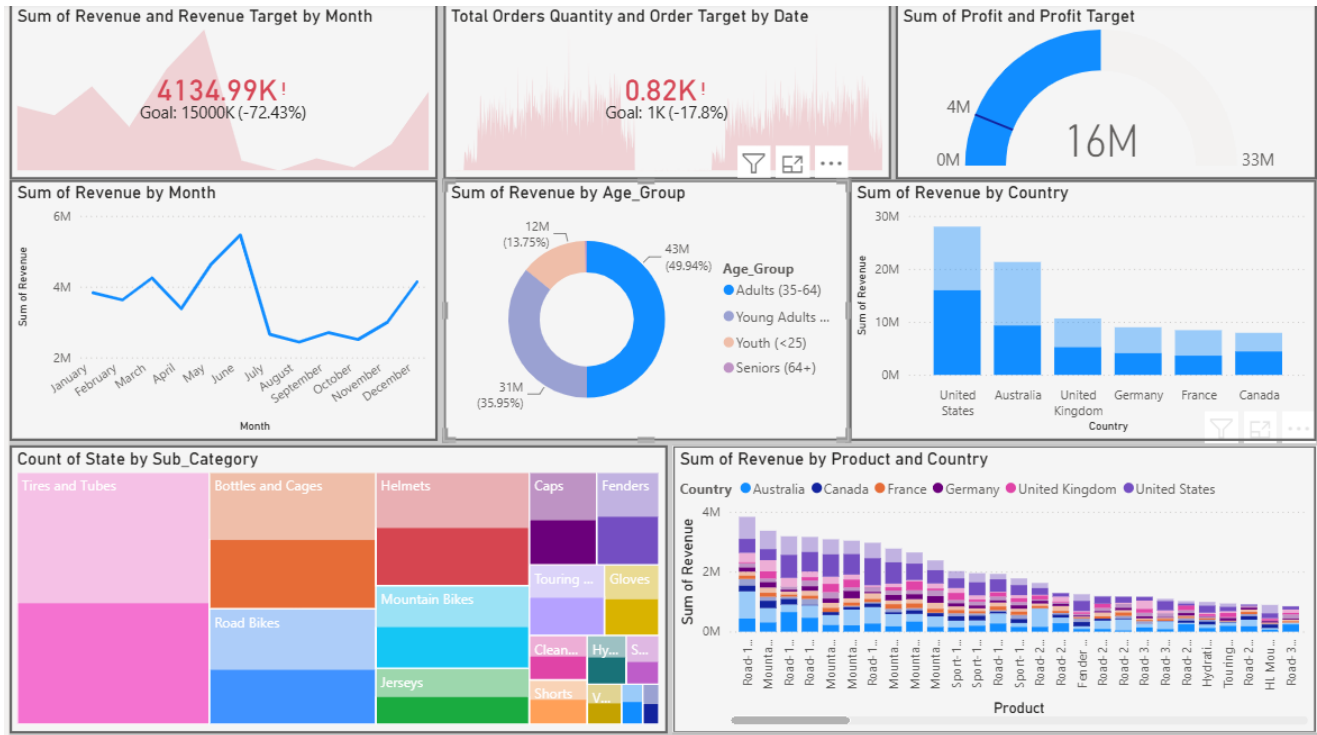


Figure 1: Sales Performance Dashboard — Power BI

Chart-by-Chart Insights

Each visual on the dashboard is chosen deliberately to answer a specific business question. Below is the context and takeaway for every chart, following the storytelling principle of pairing a visual with a clear insight rather than showing data for its own sake.

1. Revenue vs Target (Area/KPI card)

Chart type: KPI card with trend sparkline.

What it shows: Total revenue (4,134.99K) against a monthly goal of 15,000K.

Insight: Revenue is running about 72% below target. This is the single biggest red flag on the dashboard and should be the opening point of any stakeholder conversation — the target may be unrealistic for the current customer base, or there is a genuine, large revenue shortfall that needs root-cause analysis.

2. Order Quantity vs Target (KPI card)

Chart type: KPI card with trend sparkline.

What it shows: Total orders (0.82K) against a goal of 1K.

Insight: Order volume is much closer to target (-17.8%) than revenue is (-72.4%). Read together, this tells us the shortfall is driven more by average order value / pricing / product mix than by a lack of customer demand — orders are almost on pace, but each order is generating far less revenue than planned.

3. Profit vs Target (Gauge)

Chart type: Gauge chart, 0 to 33M scale.

What it shows: Current profit of 16M against a max scale of 33M.

Insight: A gauge works well here because profit is a single, target-driven KPI where the audience mainly needs to know 'are we roughly on track' at a glance. Profit sits just under the midpoint of the range, showing the business is profitable even while missing its revenue goal — a sign of reasonably healthy margins.

4. Revenue by Month (Line chart)

Chart type: Line chart, Jan-Dec.

What it shows: Monthly revenue trend across the year.

Insight: Revenue is volatile rather than steadily growing, with a sharp peak around July and a deep dip in the August-September window before partially recovering. A line chart is the right choice here because the business question is about change over time and seasonality, not comparison between discrete categories. The August/September dip is a strong candidate for deeper investigation (promotions ended, seasonality, stock-outs, etc.).

5. Revenue by Age Group (Donut chart):

Chart type: Donut chart.

What it shows: Share of revenue contributed by each customer age segment.

Insight: Adults (35-64) and Young Adults together account for the large majority of revenue (roughly 86% combined), while Youth (<25) and Seniors (64+) are minor contributors. A donut is appropriate for a small number of categories that sum to a whole; it should not be used if the segment count grows much beyond 4-5. Business takeaway: marketing and product decisions should prioritize the Adults and Young Adults segments, while Youth/Seniors represent an under-tapped growth opportunity.

6. Revenue by Country (Bar chart):

Chart type: Vertical bar chart.

What it shows: Revenue ranked by country — United States, Australia, United Kingdom, Germany, France, Canada.

Insight: A bar chart is the right tool for comparing a handful of discrete categories side by side (unlike a pie/donut, which gets harder to read past a few slices). The United States and Australia are clearly the two leading markets, together contributing noticeably more than the other four countries combined — revenue is geographically concentrated rather than evenly spread.

7. State Count by Sub-Category (Treemap):

Chart type: Treemap.

What it shows: Which product sub-categories are sold across the most states/regions.

Insight: Tires and Tubes, Road Bikes, and Bottles and Cages occupy the largest tiles, meaning they have the broadest geographic reach among states. A treemap is well suited here because it can show many categories of varying size in limited space and highlights the biggest contributors immediately through area size. Smaller, accessory-type sub-categories (caps, gloves, fenders) have a much narrower footprint.

8. Revenue by Product and Country (Stacked bar chart):

Chart type: Stacked bar chart, product on x-axis, country as stack/color.

What it shows: Revenue for each product, broken down by contributing country.

Insight: This view lets the audience see both total product performance and its country composition in one chart. Road bike and mountain bike variants lead in overall revenue, and their bars are consistently made

up of contributions from all six countries, confirming these products have broad international demand rather than being popular in a single market.

Key Business Insights:

- Revenue is significantly behind target (-72.4%) while orders are nearly on pace (-17.8%) — the shortfall is primarily a value/pricing problem, not a demand problem.
- Profit remains solid at 16M, showing decent margin discipline even in a year that missed its revenue goal.
- Two countries (United States and Australia) and two age segments (Adults 35-64 and Young Adults) drive the large majority of revenue — the business is concentrated, which is efficient but also a concentration risk.
- The July peak followed by an August-September dip in monthly revenue points to a seasonal or promotional pattern worth investigating and potentially replicating outside its natural season.
- Tires and Tubes, Road Bikes, and Bottles and Cages are the most broadly distributed and highest revenue product lines, and they sell across all six countries in the dataset.

Summary Storyboard:

The dashboard is designed to be read left-to-right, top-to-bottom, following a natural storytelling arc:

Step 1 — The headline (KPI cards + gauge)

Start with revenue, orders, and profit versus target. This immediately frames the core tension: solid profit and near-target order volume, but a large revenue shortfall.

Step 2 — The trend (line chart)

Move to the monthly revenue trend to show that the shortfall is not evenly spread across the year — there is a clear seasonal peak and dip worth explaining.

Step 3 — The audience (donut chart)

Show who is actually buying — Adults and Young Adults dominate — to connect the revenue story to a customer segment.

Step 4 — The geography (bar chart)

Show where the revenue comes from — the US and Australia lead — adding a geographic dimension to the concentration story.

Step 5 — The products (treemap + stacked bar)

Finish with what is being sold and where — Tires and Tubes, Road Bikes, and Bottles and Cages lead and sell across all markets, closing the loop from headline number down to the specific products driving it.

Design choices such as avoiding 3D effects, limiting the color palette to a consistent blue/pink/purple theme, using a gauge and cards only for single-number KPIs, and reserving the treemap and stacked bar for genuinely multi-dimensional comparisons all follow the 'right chart for the data, minimal clutter' principle from the task guidelines.